

Class 30006 – Financial Markets and Institutions
Università Commerciale Luigi Bocconi
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Final review + Exercises



Outline

- 1. BIG PICTURE: OVERVIEW OF THE COURSE**
- 2. EXERCISES (Lectures 12-22)**
- 3. EXERCISES (Lectures 1-11)**

BIG PICTURE

OVERVIEW OF THE COURSE

Overview

- ✓ The **financial system** performs the basic function of moving funds from lenders/savers to borrowers/spenders
- ✓ It uses **financial assets**, which are claims on real assets
- ✓ It can do this with
 - **direct finance** (borrowers access financial markets directly)
 - or with **indirect finance** (borrowers deal with an intermediary who is managing the ultimate lender's savings)
- ✓ The financial system also
 - reduces **transaction costs**
 - and solves two types (really? two?) of **asymmetric information**

Overview

- ✓ **Direct Finance:** direct issuance of either **debt** or **equity**
- ✓ These claims are traded in secondary markets
- ✓ Debt entails no ownership right, but it promises the debt holder a **pre-defined** (often fixed) **payment** and has a pre-defined **maturity** (although there are also perpetuities)
- ✓ Equity entails **ownership rights and has no pre-defined maturity**. Equity is the **residual claim**: its holders are entitled to whatever is left after creditors have been paid off

Overview

- ✓ Debt instruments with original maturity of less than 1 year are traded in **money markets**
- ✓ Debt instruments with original maturity of more than 1 year are traded in the **bond market** (part of **capital markets** together with the **stock market**)

Overview

- ✓ The price of any financial asset is inversely related to its return
- ✓ The **price of debt, therefore, is inversely related to interest rates** (and the longer the maturity, the more the price will fall if interest rates rise: **duration**)
- ✓ Interest rates are determined by the forces of **supply and demand** which are affected by many things (wealth, risk, $E(i)$, π^e , government deficits...)

Overview

- ✓ Equity is traded in the stock market (grouped in an index)
- ✓ There are multiple ways stocks can be traded, which are generally classified as
 - **organized exchanges (OE)**
 - and **over-the-counter markets (OTC)**
- ✓ As for any financial instrument, it is the future that matters for stock valuation, not the past
- ✓ In order to value a share of stock, one needs to discount future **expected cash flows by the appropriate discount rate**
- ✓ Equity holders expect a cash flow distribution at some point, even if the firm doesn't pay dividends

Overview

Derivatives are financial securities whose **payoffs are linked to another, previously issued security** (or asset)

We studied:

- **forwards** (OTC, not standardized, \$ exchanged in the future at delivery)
- **futures** (traded on exchanges, standardized, daily settlement)
- **options** (premium paid upfront, option value or payoff $\neq$ option profit)

An important function of derivatives is **hedging** (insurance against particular risks) but it can be risky!

Overview

- ✓ **Market efficiency** is the degree to which the **prices** of financial securities reflect those securities' **fundamental values**
- ✓ There are three forms of market efficiency:
 - ✓ **Weak**
 - ✓ **Semi- strong**
 - ✓ **Strong**
- ✓ Markets are probably semi-strong efficient most of the time, at least in the short-term
- ✓ Sometimes, however, market efficiency may break down, especially over long horizons (value investing & Mr. Buffett)

Overview

- ✓ Financial markets are plagued by **transaction costs and information asymmetries**
- ✓ These problems create a need for Financial Institutions (FIs).
 - FIs specialize in **collecting and processing information** about borrowers and financial products
 - FIs can also **minimize transaction costs**
- ✓ That's why most external finance is provided by FIs (indirect finance) rather than financial markets (direct finance)
- ✓ And **moral hazard** can explain **why debt is more common than equity** (less need for monitoring)

Overview

- ✓ **Commercial Banks** are institutions that take deposits and make private loans (and other investments)
- ✓ They need to keep **excess reserves** to manage the outflow of deposits and manage to have **enough capital** to remain solvent while giving returns to shareholders
- ✓ Their overall profitability can be assessed with **ROA**, **ROE** ($ROE = ROA \times A/E$)
- ✓ Their ability to generate cash from interest activities with the **NIM** = $\text{Net Interest Income} / A = (\text{Int. Income} - \text{Int. Exp}) / A$

Overview

- ✓ **Central banks** are in charge of monetary policy.
- ✓ Assets=securities. Liabilities=reserves+\$ in circulation.
Greatly expanded their balance sheet after 2008 (QE etc)
- ✓ They implement monetary policy (i.e. the supply of money in the economy) in the market for **bank reserves**
- ✓ The demand for reserves by banks: **interest rate paid on excess reserves** provides a floor (flattening out of curve)
- ✓ The supply of reserves by the central bank: upward vertical (monopolist) but **discount window rate** provides a ceiling

Overview

- ✓ **Mutual funds** and **hedge funds** pool money of individual investors and invest this pool into **diversified portfolios**
- ✓ These institutions help investors **minimize transaction costs** and offer **expertise**
- ✓ Main revenue: **fees**
- ✓ **ETFs** are passive open-end index funds but trade like a single stock (closed-end). ETF price=NAV through arbitrage
- ✓ MFs are regulated, while hedge funds not much

Overview

- ✓ **Investment banks** help firms issue debt and equity in primary markets (underwriting)
 - primary market transactions occur when the issuer of the security receives the funds (like IPOs)
- ✓ Company may opt for **firm commitment underwriting**, guaranteed to raise an amount, or for **best effort underwriting**
- ✓ Investment banks, but also mutual funds, and other intermediaries are prone to **conflicts of interest** (multiple objectives and those objectives conflict with each other)
- ✓ Conflict of interest is a type of **moral hazard**

Overview

- ✓ Banks are **asset transformers**: they sell liabilities (deposits) with one set of characteristic and buy assets with a different set of characteristics
- ✓ Generally, banks' liabilities are more liquid and have shorter maturities than their assets (**duration gap** > 0 , **income gap** > 0). This exposes banks to IRR
- ✓ Bank **net worth** is what is owned by **bank owners**
- ✓ Net worth (capital) is the cushion against insolvency
- ✓ Insolvency is a situation when net worth becomes negative
- ✓ Furthermore, banks need to manage **credit risk** and **interest-rate risk**

Overview

- ✓ Bank failures are costly. So the government uses a safety net, such as **deposit insurance** to prevent bank failures
- ✓ It will also rescue the too-big-to-fail banks
 - However, this can create **moral hazard** at banks
- ✓ In order to prevent excessive risk-taking, there are minimum **capital requirements** imposed on banks
- ✓ Also **risk-assessment, restrictions on asset holdings, competition**, etc..

Overview

- ✓ **Financial crises** (bank panics) occur because of uncertainty about the quality of the assets of financial institutions
- ✓ This quality may deteriorate because of sudden asset price declines, because of excessive risk-taking, or other factors
- ✓ Declines in net worth lead to **deleveraging**
- ✓ Deleveraging decreases the amount of credit; economic activity declines
- ✓ A sharp decline in prices can also lead to **debt deflation**

EXERCISES

(Lectures 12-22)

Exercise 1

Both Bank A and Bank B have total assets valued at \$5,000,000. They also have the same amount of capital, \$450,000. Bank A's risk-weighted assets are \$3,000,000 while Bank B's risk-weighted assets are \$2,500,000. Which bank is riskier?

Exercise 1 Solutions

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A: Obviously, Bank A is riskier: for the same amount of capital, it has more risk-weighted assets than bank B.

Exercise 2

A bank has a capital to asset ratio of 10% and an income gap of –\$300.

Q1: If it has \$100 of capital and 50% of its assets are rate-sensitive, what is the amount of rate-insensitive liabilities this bank has?

Exercise 2 Solutions

A bank has a capital to asset ratio of 10% and an income gap of −\$300.

Q1: If it has \$100 of capital and 50% of its assets are rate-sensitive, what is the amount of rate-inssensitive liabilities this bank has?

A1: Capital Ratio (or CR)=E/A (inverse of Equity Multiplier=A/E):

$$A = E/CR = \$100/0.1 = \$1,000$$

IncomeGap = RSA−RSL, so: $RSL = RSA - \text{IncomeGap}$

$$RSL = 50\% \times \$1,000 + \$300 = \$800$$

$$\text{Total Liab.: } L = A - E = \$1,000 - \$100 = \$900$$

Then, the amount rate-inssensitive liabilities is:

$$L - RSL = \$900 - \$800 = \$100$$

Exercise 3

Q2: Take data from previous exercise. How will banks' income change if interest rates go up by 1%? What will be the effect of this rate increase on the bank's net interest income if it converts \$100 of its rate-sensitive liabilities into rate-insensitive liabilities?

Exercise 3 Solutions

Q2: Take data from previous exercise. How will banks' income change if interest rates go up by 1%? What will be the effect of this rate increase on the bank's net interest income if it converts \$100 of its rate-sensitive liabilities into rate-inssensitive liabilities?

A2:

If RSL go down by 100 then $IncomeGap = -\$200$ (recall: $IncomeGap = RSA - RSL$), and if interest rates go up by 1%, then:

$$\Delta NII = IncomeGap \times \Delta i = -\$200 \times 0.01 = -\$2$$

Exercise 4

Bank AAA has \$400 of capital and \$4,600 in deposits.

The bank has two options.

The first option is to invest in government bonds, which yield 5% per year, risk free.

The second option is to invest in junk bonds, which with probability 60% will yield a return of 10%, and with the remaining 40% probability will yield a return of -8%.

The bank has to pay 3% per year to its depositors, regardless of the option it chooses and there are no reserve requirements.

Q: Which option will the bank owners choose?

Exercise 4 Solutions

A: Total assets are \$5,000 ($=\$4,600 + \400).

If bank invests in ...

a) govt bonds: $\$5,000 \times 5\% - \$4,600 \times 3\% = \$112$

b) junk bonds:

b1) with prob. 60%, they get $\$5,000 \times 10\% - \$4,600 \times 3\% = \$362$

b2) with prob. 40% they lose \$400, which would wipe out shareholders (all capital is depleted). In this case, the bank does not need to pay depositors any interest.

Thus, the total expected profit from investing in the junk bond is

$$\$362 \times 60\% - \$400 \times 40\% = \$57.2$$

Since $\$57.2 < \112 , the bank invests in govt bonds

Exercise 5

XYZ Bank has \$100 worth of government securities and \$150 worth of loans. It has \$250 worth of deposits. Assume that the required reserve ratio is 5%.

Q: How much reserves does this bank have if it keeps \$10 in excess reserves? How much capital does the bank have?

Exercise 5 Solutions

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Q: How much reserves does this bank have if it keeps \$10 in excess reserves? How much capital does the bank have?

A: Required Reserves = $\$250 \times 0.05 = \12.5 , and since Total Reserves (TR) = Required + Excess, then

$$TR = \$22.5 = \$10 + \$12.5$$

$$\text{Assets} = \text{Gov. securities} + \text{Loans} + \text{TR}$$

$$\text{Assets} = \$100 + \$150 + \$22.5 = \$272.5$$

$$\text{Liabilities} = \$250 \text{ (only deposits here)}$$

$$\text{Capital} = \text{Assets} - \text{Liabilities} = \$272.5 - \$250 = \$22.5$$

Exercise 6

Bank XYZ has a Net Interest Margin (NIM) of 4% and assets of \$500. It decides to borrow in the commercial paper market at 2% in order to finance a loan that pays 3%.

Q: How much did the bank borrow if its new NIM, after lending, is 3%?

Exercise 6 Solutions

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Q: How much did the bank borrow if its new NIM, after lending, is 3%?

A: Recall that $NIM = \frac{NII}{Assets} = \frac{(Interest\ Income - Interest\ Expenses)}{Assets}$.

where $NII = Net\ Interest\ Income$

Call L the new loan amount (=amount of borrowing raised with CP).

NII before new loan is $NII_0 = NIM_0 \times A_0 = 4\% \times \$500 = \$20$

After lending at 3%, the NIM shrinks from 4% to 3% which means that

$$NIM_1 = \frac{NII_0 + (i_{loans} - i_{CP}) \times L}{A_0 + L} \Rightarrow 3\% = \frac{\$20 + (3\% - 2\%) \times L}{\$500 + L}$$

Solve for L : $\$500 \times 3\% + 3\% \times L = \$20 + 1\% \times L$

$\$15 + 3\%L = \$20 + 1\%L$ or $\$5 = 2\% \times L$

or $L = \$250$

Exercise 7

An investment bank agrees to underwrite 2,000 shares of stock for firm on a best-efforts basis. The investment bank is able to sell all shares for \$10.00 per share, and it charges the firm \$0.90 per share sold. At the end of the day, the stock price is down to \$7.

Q: How much money does the firm receive? What is the gross profit to the investment bank, including fees?

Exercise 7 Solutions

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Q: How much money does the firm receive? What is the gross profit to the investment bank, including fees?

A: the firm gets $\$18,200 = 2,000 \times (\$10 - \$0.9)$...
and the IB gets $\$1,800 = 2,000 \times \0.9 .

End of day price is irrelevant.

Exercise 8

Wise Bank has assets totaling \$250 million with an average duration of 5 years, and liabilities totaling \$225 million, with an average duration of 4 years. Wise Bank is subject to Basel III requirements and therefore has to maintain a capital/asset ratio of at least 8%.

Determine if Wise Bank complies with the requirement when interest rates increase from 6% to 7%.

Exercise 8 Solutions

A:

$$\underline{t=0}: NW_0 = \$250M - \$225M = \$25M$$

$$CR_0 = \left(\frac{NW}{A} \right)_0 = \frac{\$25M}{\$250M} = 10\% > mcr (= 8\%)$$

t=1: change in capital and in assets due to Δi .

$$DUR_{GAP} = DUR - \frac{\$225M}{\$250M} 4 = 1.4$$

$$\Delta NW = \left(-DUR_{GAP} \frac{\Delta i}{1+i} \right) A = \left(-1.4 \frac{0.01}{1+0.06} \right) \$250M = -\$3.3M$$

$$\Delta A = \left(-DUR_A \frac{\Delta i}{1+i} \right) A = \left(-5 \frac{0.01}{1+0.06} \right) \$250M = -\$11.8M$$

SO, new capital ratio:

$$CR_1 = \frac{NW_1}{A_1} = \frac{NW_0 + \Delta NW}{A_0 + \Delta A} = \frac{\$25M - \$3.3M}{\$250M - \$11.8M} = 9.1\% > mcr (= 8\%)$$

Exercise 9

The CEO of firm A announces the following dividend policy from next year: \$4 per share per year and a 9% growth for ever.

The CEO of firm B announces another policy: \$120 for the next year, then \$1 for the subsequent year with a 5% growth for ever.

The required return is 12%. Which price is higher?

Exercise 9 Solutions

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Solution:

$$P(A) = \frac{Div_1}{k_e - g} = \frac{\$4}{(0.12 - 0.09)} = \frac{\$4}{0.02} = \$133.3$$

$$P(B) = \frac{\$120}{1.12} + \frac{1}{1.12} \times \frac{\$1}{(0.12 - 0.06)} = \$107.1 + \$14.9 = \$122.0$$

$$P(A) > P(B).$$

EXERCISES

(Lectures 1-11)

Exercise 10

The government is reducing the budget deficit and inflation expectations are revised upward.

Q: As a result, equilibrium (nominal) interest on long-term bonds today will: increase/decrease/unchanged/can't be ...

Exercise 10 Solutions

The government is reducing the budget deficit and inflation expectations are revised upward.

Q: As a result, equilibrium (nominal) interest on long-term bonds today will: increase/decrease/unchanged/can't be ...

A: It cannot be determined because the two move interest rates in opposite directions:

- i. the reduction in the budget deficit reduces interest rates (by decreasing supply)
- ii. $\pi^e \uparrow$ increases the interest rate (decrease of demand and increase of supply: Fisher effect)

Exercise 11

Bond A is an annual coupon bond of \$2,000 face value and of four years remaining until maturity, with 5% YTM and a market price of \$2,200. Bond B is a zero-coupon bond and has three years remaining to maturity. Then,
Which bond has shorter duration?

Exercise 11 Solutions

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Which bond has shorter duration?

Solution: The principal payment of bond A alone makes a contribution of: $\frac{\$2,000}{(1.05)^4} / \$2,200 = 74.8\%$ to its present value and therefore a contribution of $0.748 \times 4\text{yrs} = 3.0\text{ yrs}$ to its duration.

Irrespective of what the exact contributions of the remaining coupons of bond A to its overall duration are, they're positive (it's a CB!). Since duration of ZBC (bond B) is 3 years, the duration of bond A is always higher than that of bond B.

Exercise 12

Bond A is a 3-year coupon bond with a duration of 3, a face value of \$1,900 and a price of \$2,000.

Bond B is a 7-year zero coupon bond with a face value of \$1,100 and a price of \$3,000.

What is the duration of the portfolio ?

Exercise 12 Solutions

Bond A is a 3-year coupon bond with a duration of 3, a face value of \$1,900 and a price of \$2,000.

Bond B is a 7-year zero coupon bond with a face value of \$1,100 and a price of \$3,000.

What is the duration of the portfolio ?

Solution: $DUR_{Port} = DUR(A) \times w_A + DUR(B) \times w_B$

$$DUR_{Port} = 3 \times \frac{\$2,000}{\$2,000 + \$3,000} + 7 \times \frac{\$3,000}{\$2,000 + \$3,000}$$

$$DUR_{Port} = 3 \times 0.4 + 7 \times 0.6 = 5.4 \text{ years.}$$

Data on face value is irrelevant.

Exercise 13

What is the present value of a \$1,000 face value bond which pays a coupon of 7% per year in semiannual installments (i.e., two coupon payments are made every year) and matures in 1 year? Assume that payments of the bond are discounted at an annual percentage rate of $i=5\%$. (Round to 1 decimal point)

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Solution: Formula for price is $P = \sum_{t=1}^{2n} \frac{C/2}{(1+i/2)^t} + \frac{FV}{(1+i/2)^{2n}}$

First, derive $C/2$: $i_{c.r.} = \frac{C}{FV}$, so $C = \$1,000 \times 0.07 = \70 and $\frac{C}{2} = \$35$

and derive: $\frac{i}{2} = \frac{5\%}{2} = 0.025$

Then, $PV = \frac{\$35}{1.025} + \frac{\$35}{(1.025)^2} + \frac{\$1,000}{(1.025)^2} = \$1,019.3$

Exercise 14

A trader has bought copper futures at \$92 per pound. The contract consists of 10,000 pounds. She closes her position one day before delivery when the future price is \$98. On the date of delivery, the spot price turns out to be \$88. What is the profit/loss of the trader?

Exercise 14 Solutions

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Solution:

$$\pi = (\$98 - \$92) \times 10,000 = \$60,000$$

Since the position is closed before delivery, data on spot price at delivery is irrelevant.

Exercise 15

A trader has written a European call option when the spot price of the underlying asset was \$105, and the premium was \$3. The strike price of the option is \$110. At expiration, the spot price of the asset is \$125.

What was the net profit or loss of his/her counterparty?

Exercise 15 Solutions

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What was the net profit or loss of his/her counterparty?

Solution:

The trader sold a call option, so she received the premium (+\$3). Later, the price increased and so your counterparty exercised the option, so you had to pay $-\$125 + \$110 = -\$15$.

Therefore, trader's net position is a loss: $-\$15 + \$3 = -\$12$. The position of her counterparty is specular: she paid the premium but gained because the price increased: $+\$15 - \$3 = +\$12$.

The info on the price of the underlying asset is irrelevant.

Exercise 16

During the year, a firm buys spot 400,000 lbs of sugar every month to make candies.

Q1: To hedge the input price risk while minimizing counterparty risk as much as possible, the firm should buy forwards or futures?

Q2: And if each derivative contract is for the delivery of 10,000 lbs, how many contracts should the firm take?

Q3: Is it better to enter in derivatives only once at the beginning of the year or to do it every month?

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Q3: Is it better to enter in derivatives only once at the beginning of the year or to do it every month?

Solution:

A1: buy futures because they minimize the counterparty risk

A2: buy 40 futures contracts of a standard size of 10,000 lb sugar

A3: firm can only choose to buy every month because you need delivery at every month

Exercise 17

An investor has 200 shares of Stock A that sell for \$56, but to protect against the risk of a price fall, she buys 100 (European) puts with a strike of \$60 for \$4 each. At expiration the stock is selling for \$48. What is the investor's net gain or loss on his/her entire stock/option portfolio?

Exercise 17 Solutions

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Solution:

The change in the value of the investor's stock holdings due to spot prices is $\Delta V_{\text{spot}} = 200 \times (\$48 - \$56) = -\1600 .

Since the strike price is greater than the spot price at expiration, the investor exercises his/her options, for a profit of:

$$\pi^{\text{put}} = 200 \times (\$60 - \$48 - \$4) = \$1,600.$$

Thus, her net gain/loss is: $\Delta V_p = \Delta V_{\text{spot}} + \pi^{\text{put}} = -\$1,600 + \$1,600 = \0 .

Exercise 18

An investor buys a European call option on 1,000 Treasury bonds with an exercise price of \$108. If the price of the Treasury bond is \$112 at expiration, determine the premium if the profit equals \$3,500.

Solution:

Exercise 18 Solutions

An investor buys a European call option on 1,000 Treasury bonds with an exercise price of \$108. If the price of the Treasury bond is \$112 at expiration, determine the premium if the profit equals \$3,500.

Solution:

The investor exercises the call option so that she has the right to buy 1,000 Treasury bonds at \$108 and then sell them at \$112, for a profit. If the profit equaled \$3,500 this means that the profit was:

$$(\$112 - \$108) \times 1,000 - p = \$3,500.$$

$$\text{So, } p = \$4,000 - 3,500 = \$500.$$